

Simthetic ROM Accuracy & Performance Datasheet

DCDC Buck Converter (Hybrid ROM vs. MATLAB Simscape Reference)

Field	Value
Document ID	ST-DS-2026-001
Target System	Single-Phase DCDC Buck Converter
ROM Identifier	BUCK_HYBRID_MLP — version 2026-03-21-001
ROM Core Type	Hybrid: 4-state Linear State-Space + <i>mini</i> MLP residual corrector (two-stage fit)
Sample Time	50 μ s (20 kHz fixed-step)
Execution Environment	Docker container <code>simthetic/my-rom:v1</code> (Python wrapper + native compiled library)
Repository Commit	9836826

1. Executive Summary

This datasheet provides a technical evaluation of the Simthetic Hybrid Reduced Order Model (ROM) for a DCDC Buck Converter. The ROM was benchmarked against a high-fidelity **MATLAB/Simscape** reference across two stress-test scenarios (step load transient and combined line ramp + sinusoidal duty cycle modulation).

- Computational Speedup:** Simthetic executes **28× to 31× faster** than the Simscape solver.
- Output-side accuracy:** Mean Absolute Error (MAE) on the output voltage (V_{out}) stays below **1.2%** of signal range; MAE on the output current (I_{out}) stays below **2.6%** across both scenarios.
- Deployment:** The model ships as a self-contained Docker container with no MATLAB, PyTorch or GPU runtime dependencies, and the underlying ROM core compiles to a standalone C library (`.dll` / `.so`) for embedded targets.

2. Test Methodology

Reference Solver (Simscape)

- **Local solver:** Backward Euler, fixed step 5×10^{-7} s
- **Top-level Simulink solver:** Variable-step (auto), default tolerances
- **Sample time of logged signals:** $\sim 5 \mu\text{s}$ (interpolated to ROM grid for comparison)

Simthetic ROM

- **Container:** `simthetic/my-rom:v1`
- **Invocation:** Single REST call per scenario via the on-host MCP / REST bridge
- **Reported execution time:** ROM-internal `elapsed_s` from `sim.json` (does **not** include container startup or HTTP round-trip)

Error Metrics

For each signal, the Simscape trace is linearly interpolated onto the ROM time grid and compared sample-by-sample. All percentages are normalised by the **signal's peak-to-peak range over the scenario** (**max** — **min** of the reference). For approximately-constant signals (e.g. V_{in} on Vector 1), the **nominal magnitude** is used as the normaliser instead.

Symbol	Definition
MAE %	$\frac{1}{N} \sum e_k / \text{range} \cdot 100$ — average absolute error
RMSE %	$\sqrt{\frac{1}{N} \sum e_k^2} / \text{range} \cdot 100$ — root-mean-square error
MaxErr %	$\max_k e_k / \text{range} \cdot 100$ — single worst-sample deviation

3. ROM Operating Range

The ROM was trained over the following parameter envelope. Behaviour outside this envelope is undefined; for new operating regions Simthetic recommends a re-fit.

Channel	Min	Max	Units
$V_{in,set}$ (<code>Vin_int</code>)	10	45	V
R_{load} (<code>Rload_int</code>)	1	10	Ω
Duty cycle (<code>DC_int</code>)	0.2	0.8	—

Both benchmark scenarios below sit **fully inside** this range, including the lower-boundary case ($R_{load} = 1\ \Omega$ in Vector 1).

4. Quantitative Accuracy

Vector 1 — Step Load Transient

Constant $V_{in} = 24\text{ V}$, $DC = 0.5$. Load steps: $10\ \Omega \rightarrow 1\ \Omega$ at $t = 25\text{ ms}$, $1\ \Omega \rightarrow 10\ \Omega$ at $t = 100\text{ ms}$. Total duration **250 ms**.

Signal	MAE %	RMSE %	MaxErr %	Reference Range
V_{out}	0.52%	2.24%	58.20% ¹	36.25 V
I_{out}	1.07%	2.01%	37.10% ¹	11.88 A
V_{in}	0.86%	0.99%	10.20%	24.0 V (nominal)
I_{in}	4.75%	11.52%	285.08% ²	6.19 A

Vector 2 — Line Ramp + Sinusoidal Duty Cycle

V_{in} ramp $20\text{ V} \rightarrow 40\text{ V}$ over the first **100 ms**, then held.
 $DC = 0.45 + 0.05 \sin(2\pi \cdot 100\text{ Hz} \cdot t)$. $R_{load} = 5\ \Omega$ constant. Total duration **250 ms**.

Signal	MAE %	RMSE %	MaxErr %	Reference Range
V_{out}	1.17%	1.68%	26.84% ¹	19.95 V
I_{out}	2.51%	3.02%	25.50% ¹	3.99 A
V_{in}	0.28%	0.45%	1.79%	20.00 V
I_{in}	8.20%	23.65%	680.05% ²	2.07 A

Notes on MaxErr % Spikes

¹ **Switching-instant phase offset** — At a step transition or sharp control event, the ROM produces its next sample one ROM-period (**50 μs**) after the Simscape solver has already executed several sub-microsecond integration steps. The resulting two-trace misalignment produces a single-sample percent error that is large at the instant of the event but decays to <2 % within a few hundred microseconds. As shown in the §5 waveform plots, the post-transient settling is in agreement with the reference.

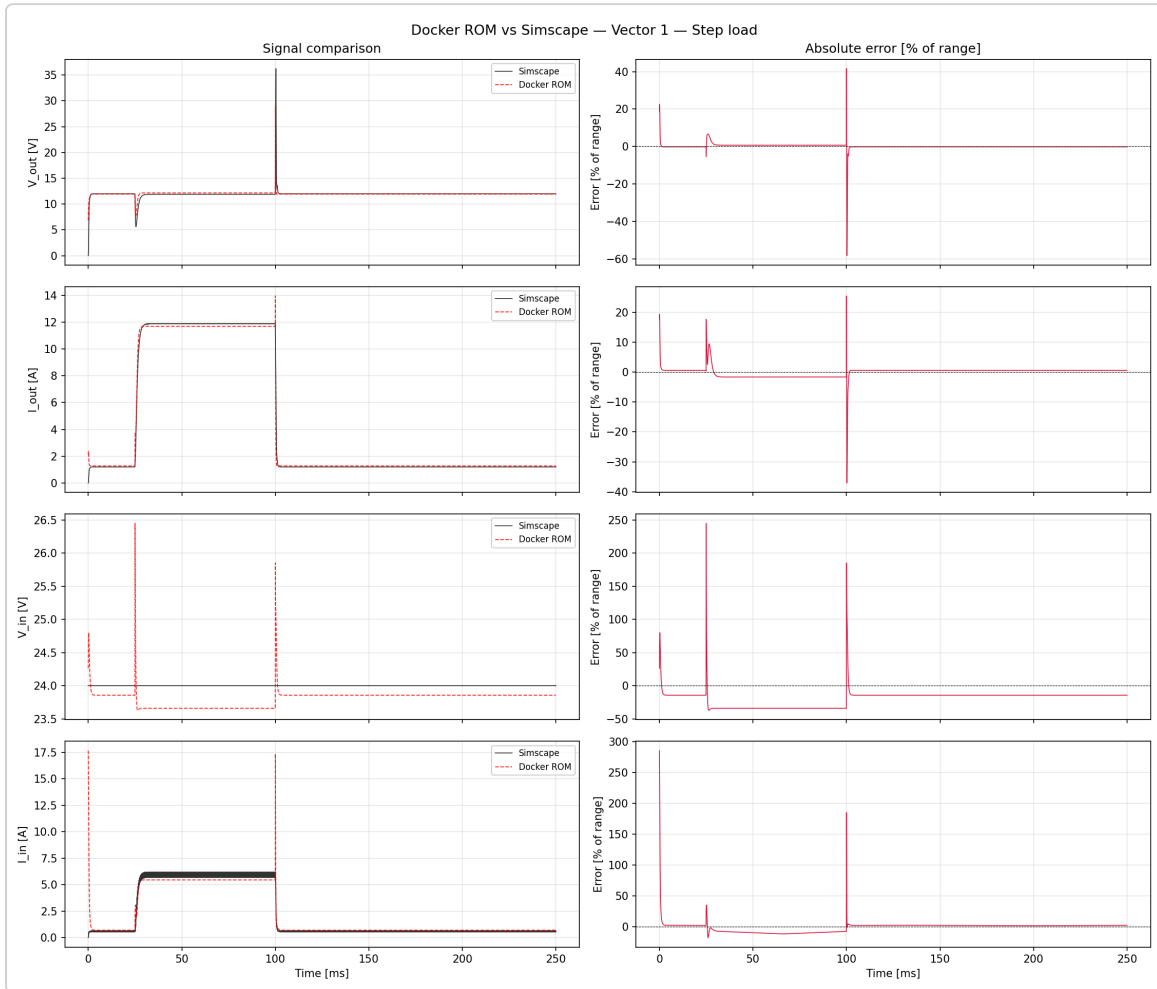
² **Input current (I_{in}) — small-range normalisation** — I_{in} has the smallest peak-to-peak range of all four signals (**6.19 A** for V1 and **2.07 A** for V2). Because the percentage error divides by that range, the same absolute error magnitude that produces 2–3 % on V_{out} produces a much larger number here. The absolute MAE on I_{in} is

0.29 A (V1) and 0.17 A (V2). Customers using I_{in} for power-balance or efficiency calculations should evaluate this against their target tolerance.

5. Visual Waveform Validation

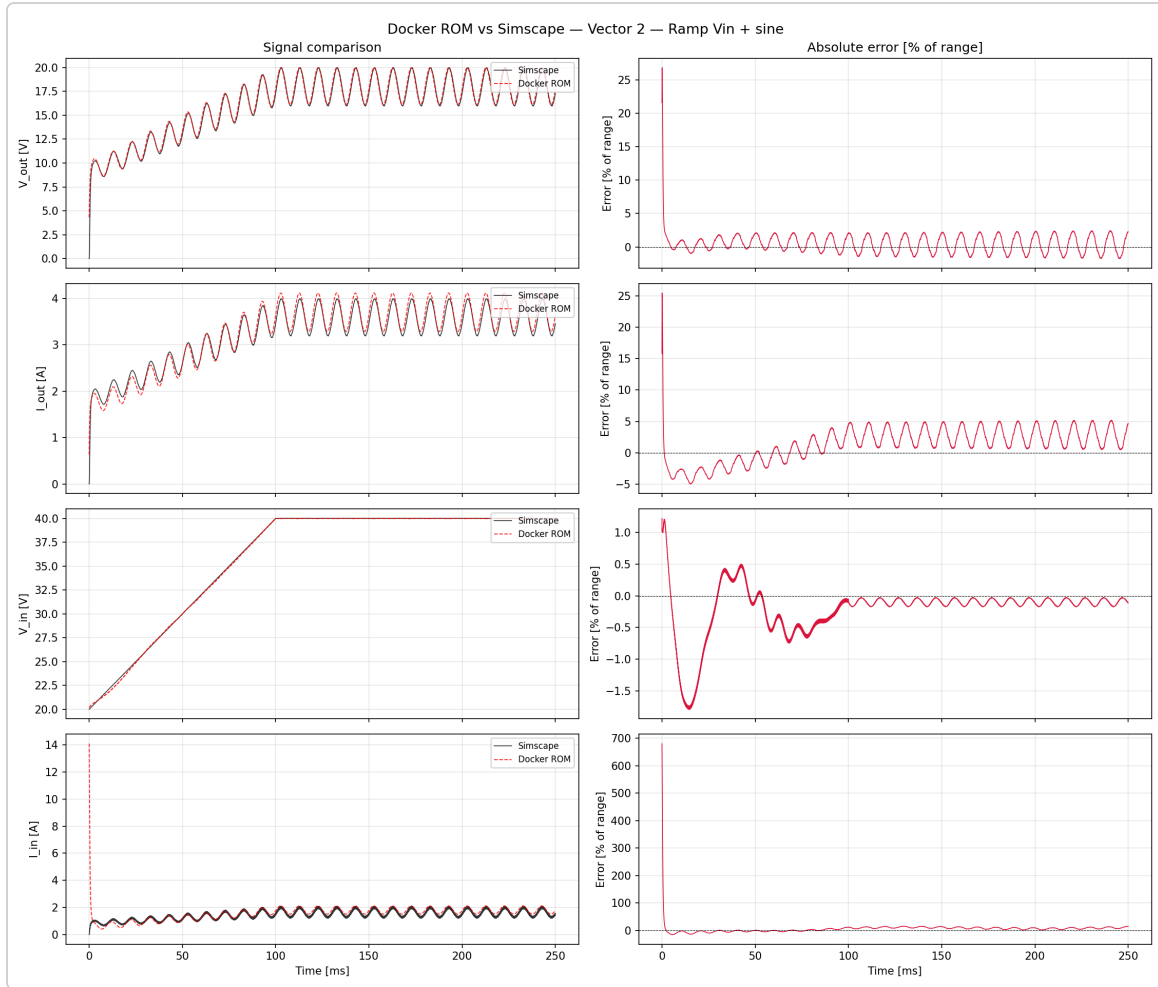
The plots below compare the transient trajectories of the MATLAB/Simscape reference (solid black) and the Simthetic ROM (dashed red). The right column shows the per-sample error as a percentage of the reference signal range.

Vector 1 — Step Load Response



- V_{out} dynamics: the ROM captures the inductive voltage dip and recovery during the $10\Omega \rightarrow 1\Omega$ load step.
- I_{out} step: the current profile matches the reference with **1.07% MAE**.

Vector 2 — Ramp V_{in} + Sinusoidal Duty Cycle



- Under simultaneous line voltage ramping ($20\text{ V} \rightarrow 40\text{ V}$) and 100 Hz duty cycle perturbation, the ROM tracks both envelope and phase of V_{out} and I_{out} .

6. Computational Efficiency

Scenario	Physical Duration	Simscape Wall-clock	Simthetic ROM	Speedup
Vector 1	250 ms	23.38 s	765.4 ms	30.6×
Vector 2	250 ms	22.61 s	818.7 ms	27.6×

Notes - ROM time is the internal model execution time reported by the container; it does **not** include container startup (~1 s, one-time cost amortised across many runs) or HTTP overhead. - Simscape was run on the same host under the solver settings in §2. - The native C-library underlying the ROM can be linked directly into a host application (FFI, FMU, or embedded target) to remove the Docker/HTTP layer entirely.

7. Deployment Options

- **Docker REST API** (current benchmark setup): zero-install on Linux/Windows/macOS, language-agnostic via HTTP, no MATLAB or PyTorch licences required.
- **Embedded C library** (`.dll` / `.so`): the ROM core compiles to a self-contained library suitable for direct linkage in C/C++ host code or HIL platforms.
- **FMU export**: FMI-compliant package for integration with Simulink, Dymola, OpenModelica and similar tools.

8. Known Limitations

- **Operating envelope**: accuracy outside the ranges in §3 is not validated. The current ROM was *not* trained on transients that take any input variable across the boundary mid-simulation.
- **I_{in} transient accuracy**: as detailed in §4 note ², the input-current trace shows larger relative deviation than the output-side signals, particularly at switching instants. Applications that close a control loop on I_{in} should review the absolute error magnitudes for fit.
- **Single-phase topology only**: this ROM does not model multi-phase or interleaved Buck variants.

Contact & Demo Scheduling

To explore a custom pilot engagement using your own simulation topology, contact:

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